Build Trovald's rig
The same machine used by the creator of Linux, including the power supply, cabineet and fans. **https://dgit.in/jun20-110**

Scorn demo using RTX
The gameplay demo from Scorn, meant to show off the next gen XBox was running on an RTX card. **https://dgit.in/jun20-111**

Read this before your next hard drive purchase

Agent001 gets a little preachy about hard drives this month and it's for your own good

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Hard drives are certainly doing a lot to tear off the "commodity" label and some of the innovations we've seen over the course of last year seem to indicate that they might actually be making a significant leap in terms of performance and giving cheap QLC SSDs a fight for inclusion in your next PC. However, when you take two steps forward, your competition is going to try to get two steps ahead. Or you could make their lives easier and take two steps back. It seems all hard drive manufacturers decided to collectively take a few steps back.

Earlier this year, a few keen-eyed individuals noticed that certain NAS drives were taking too much time to rebuild a storage pool after getting a fresh replacement drive. A few minutes here and there with the build time is normal but when you add a few days to a task that usually takes 12-16 hours, then you know something is seriously off. Turns out that some of the hard drives in the lineup were replaced with a different technology that suffers a major drop in performance under specific conditions. And they did it without labelling or informing consumers about the drastic change in specs.

Traditionally, hard drives use a technology called CMR or Conventional Magnetic Recording. In hard drives, the poles of the read/write heads are aligned perpendicular to the hard drive surface. The hard drive surface in CMR drives have tiny magnetic bits which are aligned in a certain manner to store data. These bits do not overlap one another so they take up plenty of space on the hard drive platter. In SMR or Shingled Magnetic Recording, these bits have a slight overlap over one another, thus, allowing more magnetic bits to be placed in the same surface area. Because of this, SMR drives offer much higher capacity than CMR but since SMR has overlaps, writing to these hard drives isn't so plain and simple. It takes a lot more time. However, most consumers will not realise this difference. Hard drives have had something called a buffer which is very fast and stores whatever data you want to write onto the drive temporarily. The buffer then slowly releases data to be written directly onto the hard drive at a comfortable pace. As long as the buffer has capacity, you will not notice any drop in speeds but if the buffer overflows and you have to write directly to the disk, then the transfer speeds take a massive hit. That's what happened.

When you have bursty write operations, the buffer handles everything like a champ but with continuous write operations, the buffer will hit capacity and you will slow down. NAS rebuilding operations are exactly like that. The fresh drive reads from all the hard drives in the pool, checks parity against the parity sliver and then rebuilds itself. This is very intensive and SMR drives suffer from a buffer overflow instantly. This is why NAS

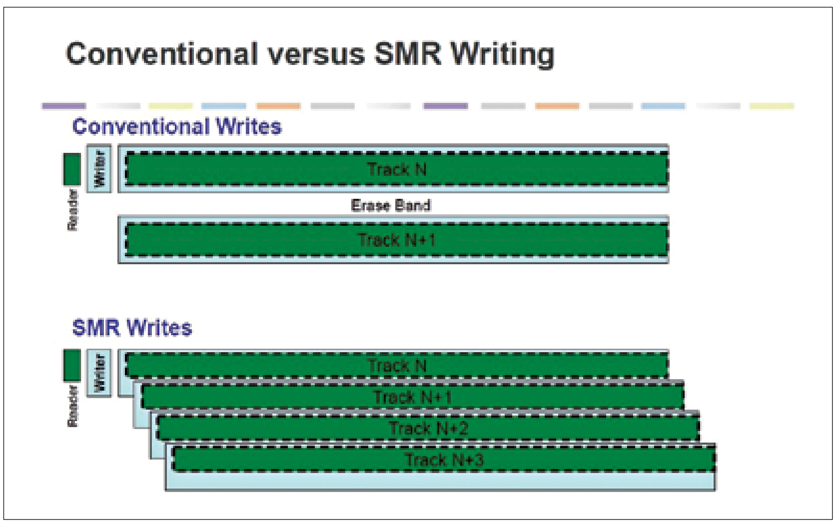

Image courtesy: **SNIA**

CMR vs SMR